

Database Requirement Analysis – Attendance Management System

Overview

The Attendance Management System requires a relational database to store and manage information related to students, faculty, subjects, attendance records, and administrators. The database is designed to ensure data consistency, efficient retrieval, and secure management of attendance information.

Database Tables

1. Admin Table

- admin_id (PK, Auto Increment)
- username (Unique, Not Null)
- password (Not Null)

2. Student Table

- student_id (PK, Auto Increment)
- register_no (Unique, Not Null)
- name
- email
- department
- year
- password

3. Faculty Table

- faculty_id (PK, Auto Increment)
- name
- email
- department
- password

4. Subject Table

- subject_id (PK, Auto Increment)
- subject_code (Unique)
- subject_name
- faculty_id (FK)

5. Attendance Table

- attendance_id (PK, Auto Increment)
- student_id (FK)
- subject_id (FK)
- attendance_date
- status (Present/Absent)

6. QR_Attendance Table

- qr_id (PK, Auto Increment)
- subject_id (FK)
- generated_time
- expiry_time

Database Relationships

- Faculty (1:M) Subject

- Student (1:M) Attendance
- Subject (1:M) Attendance
- Subject (1:M) QR_Attendance

Deliverable:

- Database Table List
- Attributes Definition
- Primary Keys
- Foreign Keys
- Relationship Mapping